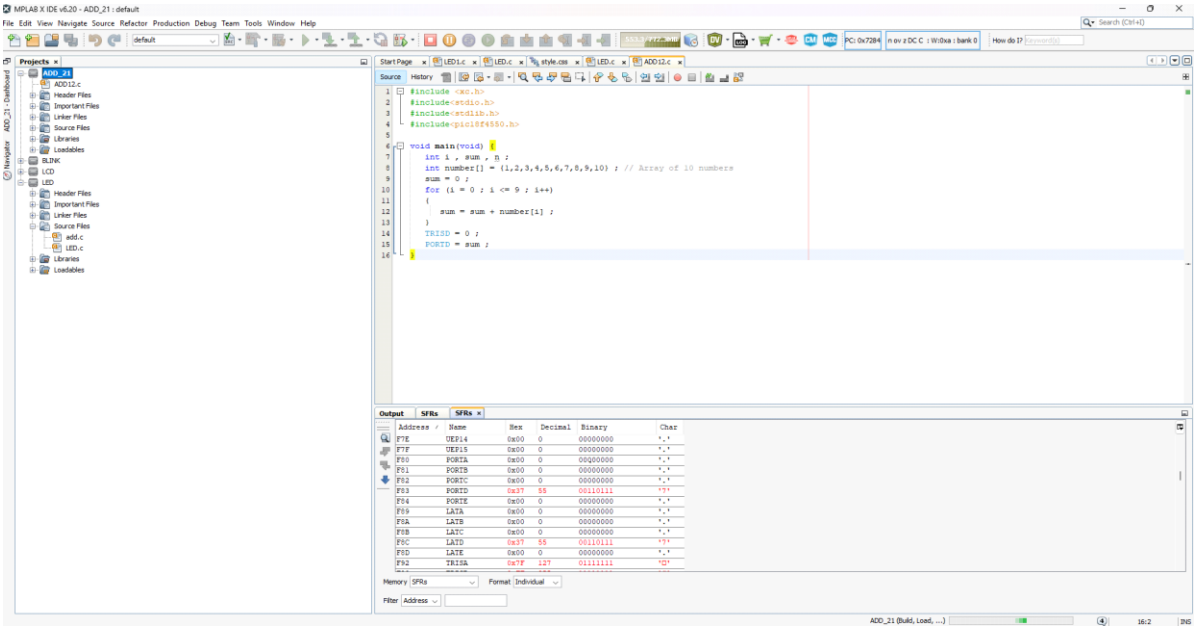
 # Program:-1(ADDITION OF ARRAY)

```
#include <xc.h>

#include<stdio.h>

#include<stdlib.h>

#include<p18f4550.h>


void main(void) {

    int i , sum , n ;

    int number[] = {1,2,3,4,5,6,7,8,9,10} ; // Array of 10 numbers

    sum = 0 ;

    for (i = 0 ; i <= 9 ; i++)

    {

        sum = sum + number[i] ;

    }

    TRISD = 0 ;

    PORTD = sum ;

}
```

OUTPUT:-


MPLAB X IDE v6.20 - ADD_21 : default
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BLINK
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LED1.c
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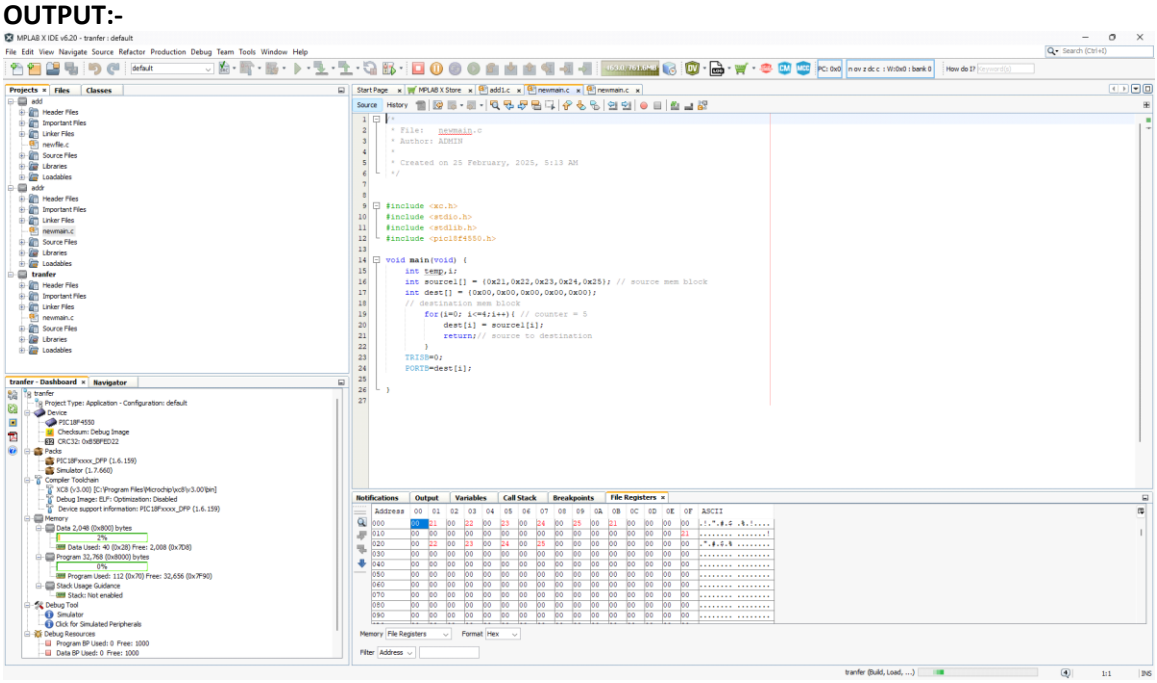
Source
History
#include <xc.h>
#include<stdio.h>
#include<stdlib.h>
#include<pic18f4550.h>
void main(void) {
int i , sum , n ;
int number[] = {1,2,3,4,5,6,7,8,9,10} ; // Array of 10 numbers
sum = 0 ;
for (i = 0 ; i <= 9 ; i++)
{
sum = sum + number[i] ;
}
TRISD = 0 ;
PORTD = sum ;
}
Output
SFRs
Address
Name
Hex
Decimal
Binary
Char
F7E UEP14 0x00 0 00000000 '.'
F7F UEP15 0x00 0 00000000 '.'
F80 PORTA 0x00 0 00000000 '.'
F81 PORTB 0x00 0 00000000 '.'
F82 PORTC 0x00 0 00000000 '.'
F83 PORTD 0x37 55 00110111 '7'
F84 PORTE 0x00 0 00000000 '.'
F89 LATA 0x00 0 00000000 '.'
F8A LATB 0x00 0 00000000 '.'
F8B LATC 0x00 0 00000000 '.'
F8C LATD 0x37 55 00110111 '7'
F8D LATE 0x00 0 00000000 '.'
F92 TRISA 0x7F 127 01111111 'D'
Memory SFRs
Format Individual
Filter Address
ADD_21 (Build, Load, ...)
16:2
INS

PROGRAM:-2 (INTERNAL TO INTERNAL MEMORY TR)

```
#include <stdio.h>

#include <stdlib.h>

#include<pic18f4550.h>


void main(main){

    int temp , i ;

    int source[] = {0x21,0x22,0x23,0x24,0x25} ; // Source memory block

    int dest[] = {0x00,0x00,0x00,0x00,0x00} ; // Destination memory block

    for(i = 0 ; i <= 5 ; i++)

    {

        dest[i] = source[i] ;

    }

  // OUTPUT USING PORT IF NEEDED

}
```

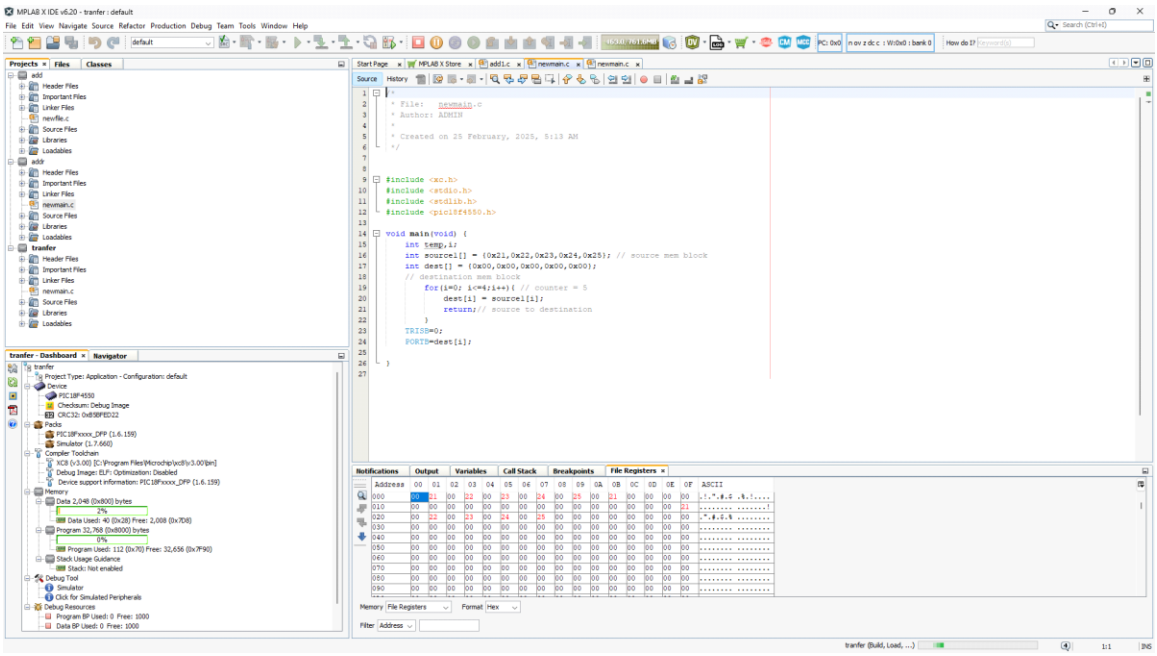
OUTPUT:-



PROGRAM:-3(MEMORY TRANSFER INTERNAL TO EX)

```
#include <stdio.h>
#include <stdlib.h>

void main(void) {
    int temp , i ;
    int source[] = {0x21,0x22,0x23,0x24,0x25} ; // Source memory block
    int dest[] = {0x99,0x98,0x97,0x96,0x95} ; // Destination memory block
    for(i = 0 ; i < 5 ; i++)
    {
        temp = source[i] ;
        source[i] = dest[i] ;
        dest[i] = temp ;
    }
    // OUTPUT USING PORT IF NEEDED
}
```

OUTPUT:-


MPLAB X IDE v6.20 - tranfer : default
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addr
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tranfer
tranfer - Dashboard
Navigator
Project Type: Application - Configuration: default
Device
PIC18F4550
Checksum: Debug Image
CRC32: 0x858FED22
Packs
Simulator (1.7.660)
Compiler Toolchain
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Debug Resources
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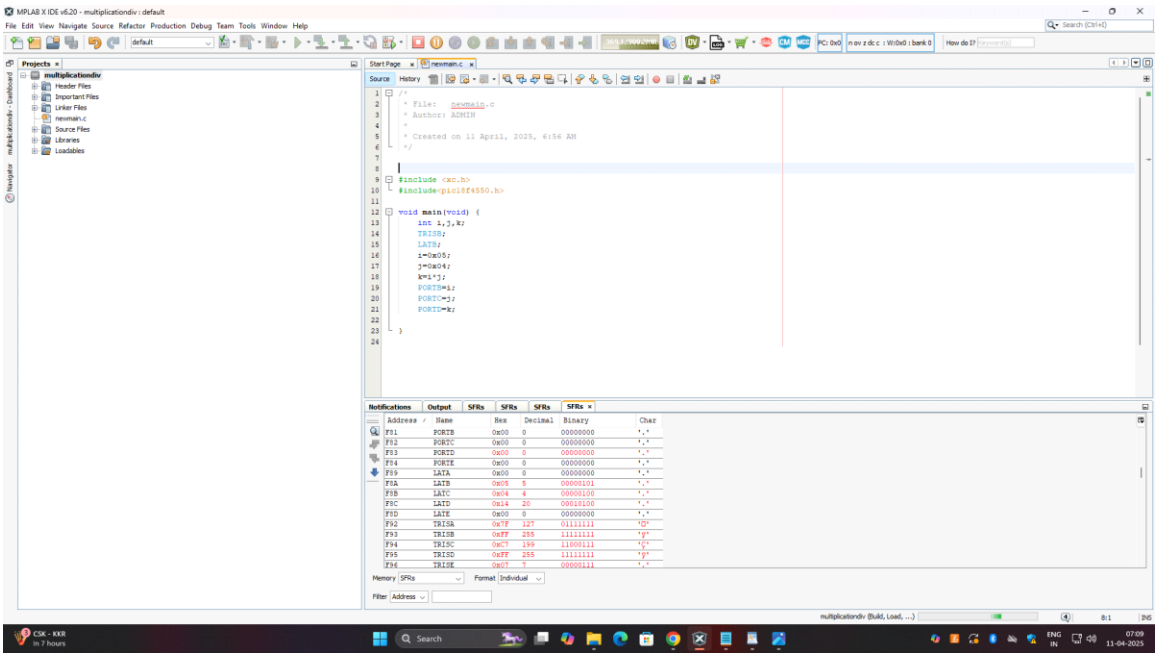
add1.c
newmain.c
Source
History
* File: newmain.c
* Author: ADMIN
* Created on 25 February, 2025, 5:13 AM
#include <xc.h>
#include <stdio.h>
#include <stdlib.h>
#include <pic18f4550.h>
void main(void) {
int temp,i;
int source1[] = {0x21,0x22,0x23,0x24,0x25}; // source mem block
int dest[] = {0x00,0x00,0x00,0x00,0x00};
// destination mem block
for(i=0; i<=4;i++){ // counter = 5
dest[i] = source1[i];
return;// source to destination
}
TRISB=0;
PORTB=dest[i];
}
Notifications
Output
Variables
Call Stack
Breakpoints
File Registers
Address
ASCII
Memory
File Registers
Format
Hex
Filter
Address

PROGRAM:- 4(MULTIPLACTION /DIVISION)

```
#include <stdio.h>

#include <stdlib.h>

#include<pic18f4550.h>


void main(void) {

    int num1 = 0x06 , num2 = 0x02 ;

    int d ;

    d = 0 ;

    d = num1/num2 ;

    TRISD = 0 ;

    PORTD = d ;

    int n1 , n2 ;

    n1 = 0x23 , n2 = 0x10 ;

    int prod = 0 ;

    for (int i = 1 ; i <= n2 ; i++){

        prod = prod + n1 ;

    } // OR prod = n1*n2

    TRISB = 0 ;

    PORTB = prod ;

}
```

OUTPUT:


MPLAB X IDE v6.20 - multiplicationdiv : default
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Projects
multiplicationdiv
Header Files
Important Files
Linker Files
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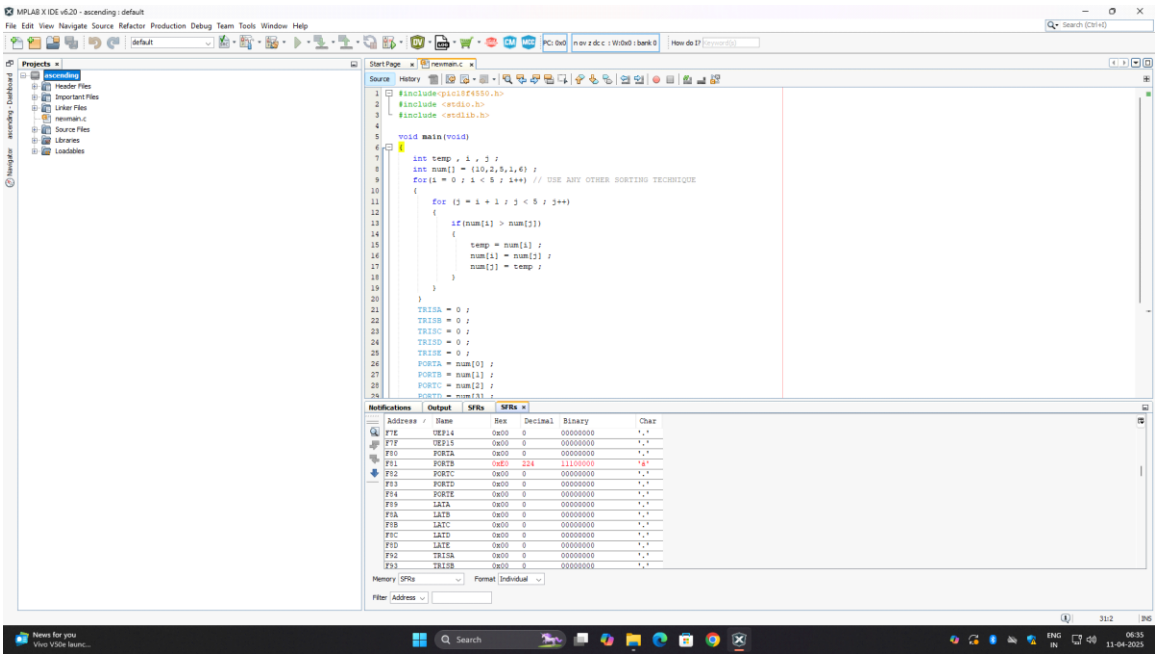
Source Files
Libraries
Loadables
Start Page
newmain.c
Source History
/*
* File: newmain.c
* Author: ADMIN
*
* Created on 11 April, 2025, 6:56 AM
*/
#include <xc.h>
#include<pic18f4550.h>
void main(void) {
int i,j,k;
TRISB;
LATB;
i=0x05;
j=0x04;
k=i*j;
PORTB=i;
PORTC=j;
PORTD=k;
}
Notifications
Output
SFRs
Address
Name
Hex
Decimal
Binary
Char
F81 PORTB 0x00 0 00000000 '.'
F82 PORTC 0x00 0 00000000 '.'
F83 PORTD 0x00 0 00000000 '.'
F84 PORTE 0x00 0 00000000 '.'
F89 LATA 0x00 0 00000000 '.'
F8A LATB 0x05 5 00000101 '.'
F8B LATC 0x04 4 00000100 '.'
F8C LATD 0x14 20 00010100 '.'
F8D LATE 0x00 0 00000000 '.'
F92 TRISA 0x7F 127 01111111 'O'
F93 TRISB 0xFF 255 11111111 'y'
F94 TRISC 0xC7 199 11000111 'G'
F95 TRISD 0xFF 255 11111111 'y'
F96 TRISE 0x07 7 00000111 '.'
Memory SFRs
Format Individual
Filter Address
multiplicationdiv (Build, Load, ...)
8:1
INS
CSK - KKR
in 7 hours
Search
ENG
IN
07:09
11-04-2025

PROGRAM:-5(ASCENDING)

```
#include<pic18f4550.h>

#include <stdio.h>

#include <stdlib.h>


void main(void)

{

    int temp , i , j ;

    int num[] = {10,2,5,1,6} ;

    for(i = 0 ; i < 5 ; i++) // USE ANY OTHER SORTING TECHNIQUE

    {

        for (j = i + 1 ; j < 5 ; j++)

        {

            if(num[i] > num[j])

            {

                temp = num[i] ;

                num[i] = num[j] ;

                num[j] = temp ;

            }

        }

     }

     TRISA = 0 ;

     TRISB = 0 ;

     TRISC = 0 ;

     TRISD = 0 ;

     TRISE = 0 ;

     PORTA = num[0] ;

     PORTB = num[1] ;
```

```
    PORTC = num[2] ;

    PORTD = num[3] ;

    PORTE = num[4] ;

}
```

OUTPUT:-

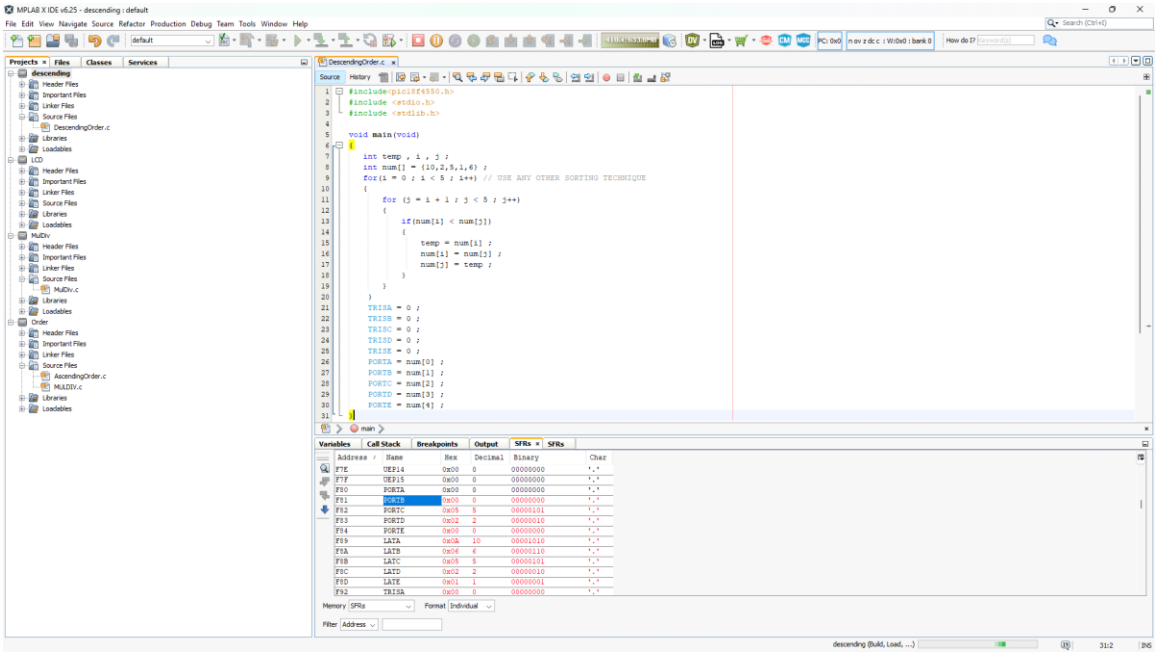


PROGRAM:-6(DESCENDING)

```
#include<pic18f4550.h>
#include <stdio.h>
#include <stdlib.h>

void main(void)
{
    int temp , i , j ;
    int num[] = {10,2,5,1,6} ;
    for(i = 0 ; i < 5 ; i++) // USE ANY OTHER SORTING TECHNIQUE
    {
        for (j = i + 1 ; j < 5 ; j++)
        {
            if(num[i] < num[j])
            {
                temp = num[i] ;
                num[i] = num[j] ;
                num[j] = temp ;
            }
        }
    }
    TRISA = 0 ;
    TRISB = 0 ;
    TRISC = 0 ;
    TRISD = 0 ;
    TRISE = 0 ;
```

```
    PORTA = num[0] ;

    PORTB = num[1] ;

    PORTC = num[2] ;

    PORTD = num[3] ;

    PORTE = num[4] ;

}
```

OUTPUT:-

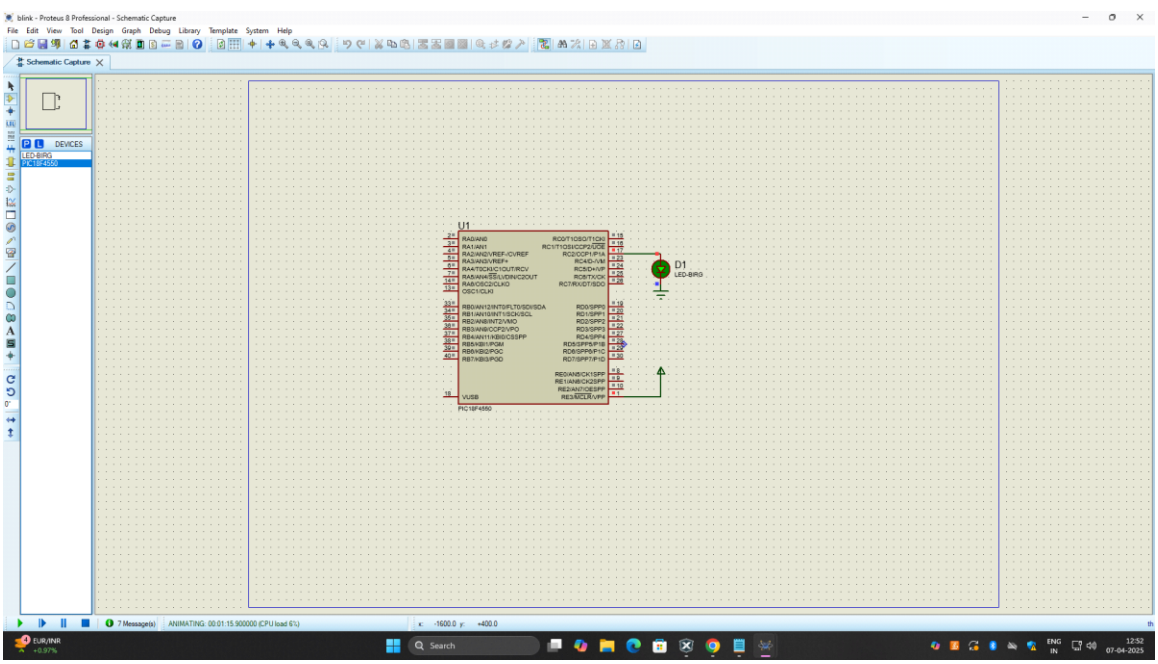


PROGRAM:-7(LED)

```
#include <xc.h>
  // Include Controller specific .h

// Configuration bit settings
#pragma config FOSC = HS            // Oscillator Selection
#pragma config WDT = OFF            // Disable Watchdog timer
#pragma config LVP = OFF            // Disable Low Voltage Programming

// Declarations
#define LED PORTCbits.RC2           // LED interfaced to RC2 pin of microcontroller

// Function Prototypes
void msdelay(unsigned int time); // Function for delay

// Start of Program Code
void main() // Main Program
{
        TRISCbits.TRISC2 = 0; // Configure RC2 (LED Pin) as output
        LED = 0; // Turn OFF LED at initial stage

        while (1)
        {
                LED = 1; // Turn ON the LED
                msdelay(250); // Delay for 250ms
                LED = 0; // Turn OFF the LED
                msdelay(250); // Delay for 250ms
```

```
    }
}

// Function Definitions
void msdelay(unsigned int time) // Function for delay
{
    unsigned int i, j;
    for (i = 0; i < time; i++)        // Outer loop for time
        for (j = 0; j < 275; j++); // Inner loop for ~1 ms delay (calibrated)
}
```

OUTPUT:-



PROGRAM:-8(LCD)

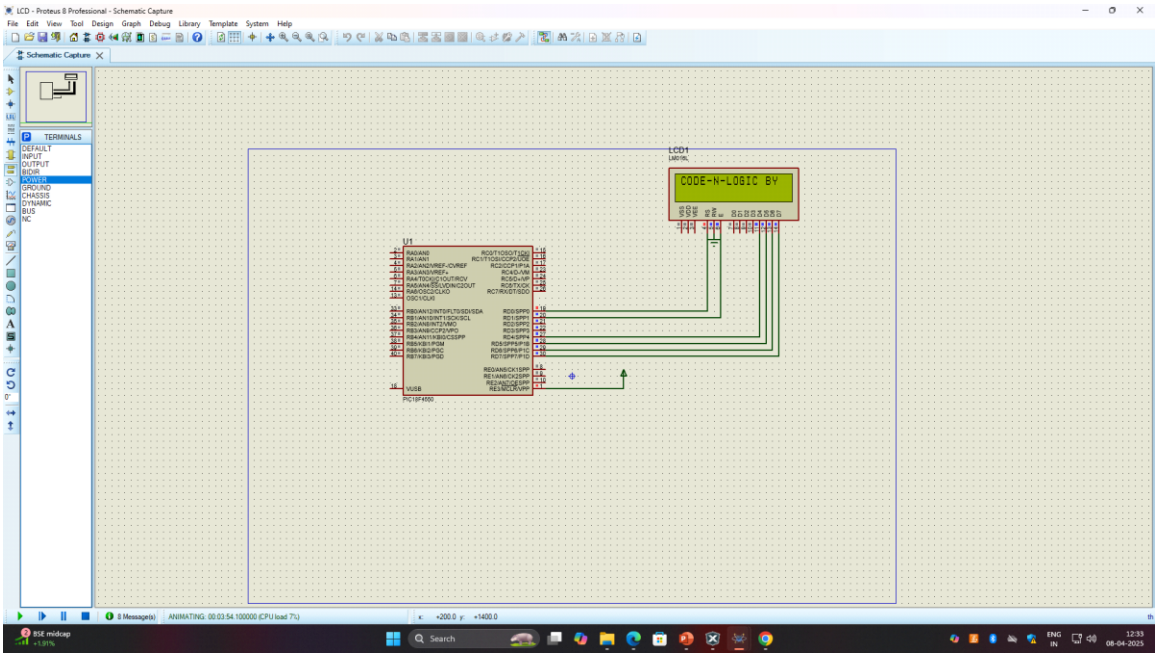
```
#include <xc.h>    // Include the correct header for the compiler

#define lcd PORTD                    //----Assigning PORTD as lcd
#define rs PORTDbits.RD0        //----rs pin for LCD control
#define en PORTDbits.RD1        //----en pin for LCD control

//------Prototype Declarations------//
void lcd_cmd(unsigned char x);            //----function to send command to lcd
void lcd_cmd_hf(unsigned char x);      //----function to send 4-bit command to lcd
void lcd_dwr(unsigned char x);            //----function to send data to lcd
void lcd_msg(char *c);                        //----function to send string of data to lcd
void lcd_init();                                //----lcd initialization
void lcd_lat();                                  //----function to latch lcd data

void delay(unsigned int ms);                //----Delay function
//----------------------------------//

void main()
{
      TRISD = 0x00;                            //---PORTD is Output
      lcd_init();                              //---LCD Initialization
      lcd_msg("CODE-N-LOGIC BY");    //---Send String
      lcd_cmd(0xC0);                           //---Command for next line in LCD
      lcd_msg("SCOE & GROUP");    //---Send String
      while(1);                                //---Wait here
}
```

```
//-----End of Program-------//

void lcd_lat()
{
	//---Latching function (high to low)
	en = 1;		//----Enable Pin is high
	delay(1);	//----1ms delay
	en = 0;		//----Enable Pin is Low
}

void lcd_cmd(unsigned char x)
{
	rs = 0;				//----Register Selected is Command register
	lcd &= 0x0F;			//----Masking higher 4-bits of PORTD
	lcd |= (x & 0xF0); //----Masking lower 4-bits of Command
	lcd_lat();			//----Latching it to lcd

	lcd &= 0x0F;			//----Masking higher 4-bits of PORTD
	lcd |= ((x & 0x0F)<<4); //----Masking lower 4-bits of Command
	lcd_lat();			//----Latching it to lcd
}

void lcd_cmd_hf(unsigned char x)
{
	rs = 0;				//----Register Selected is Command register
	lcd &= 0x0F;			//----Masking higher 4-bits of PORTD
	lcd |= (x & 0xF0); //----Masking lower 4-bits of Command
	lcd_lat();			//----Latching it to lcd
```

```
}

void lcd_dwr(unsigned char x)
{
	rs = 1;                         //----Register Select Pin is set to Data Register
	lcd &= 0x0F;                    //----Masking higher 4-bits of PORTD
	lcd |= ((x & 0xF0)); //----Masking lower 4-bits of Data
	lcd_lat();                      //----Latching it to lcd

	lcd &= 0x0F;                    //----Masking higher 4-bits of PORTD
	lcd |= ((x & 0x0F)<<4); //----Masking lower 4-bits of Data
	lcd_lat();                      //----Latching it to lcd
}

void lcd_msg(char *c)    // Change unsigned char* to char* to match string type
{
	while(*c != 0)
	{
		lcd_dwr(*c++);
	}
}

void delay(unsigned int ms)
{
	int i,j;
	for(i=0;i<=ms;i++)
		for(j=0;j<=120;j++);    // Small delay
}
```

```
void lcd_init()
{
        lcd_cmd_hf(0x30);      //---Sending Sequence Command
        lcd_cmd_hf(0x20);      //---Sending Sequence Command
        lcd_cmd(0x28);         //---Command to Select 4-bit LCD
        lcd_cmd(0x0E);         //---Cursor Blinking
        lcd_cmd(0x01);         //---Clear LCD Display
        lcd_cmd(0x06);         //---Auto-Increment LCD
        lcd_cmd(0x80);         //---Location address
}
```

OUTPUT:-

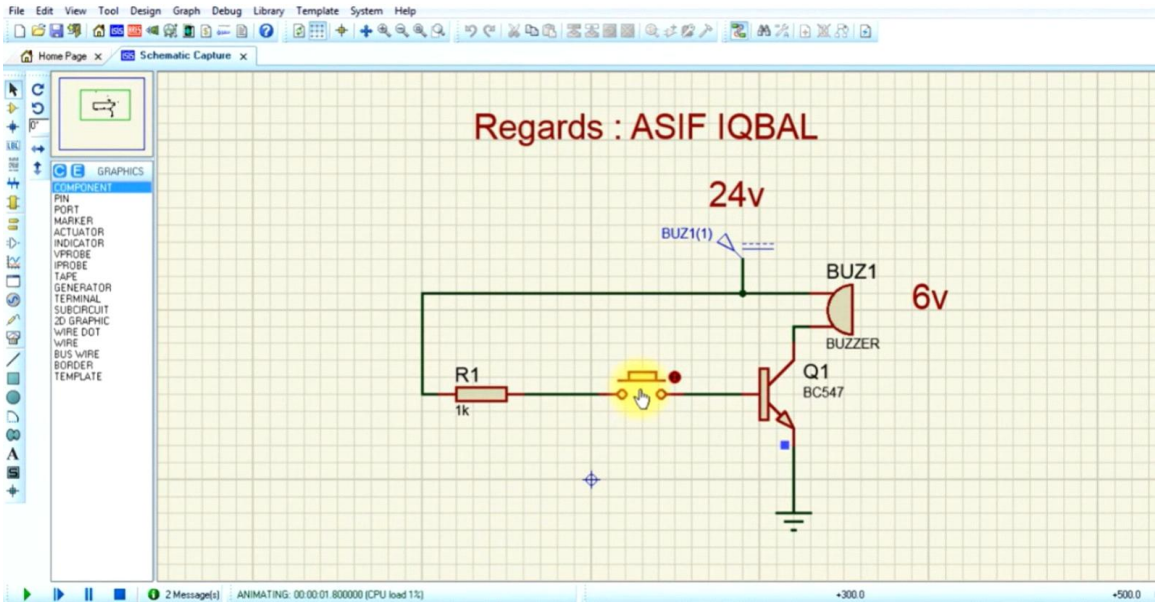


PROGRAM:-9(BUZZER)

```
#include <pic18f4550.h>
/* Contains PIC18F4550
specifications */
#define Buzzer LATAbits.LATA5
/* Define buzzer pin */
unsigned int count = 0;
void interrupt Timer1_ISR()
{
if(TMR1IF==1)
{
//1 ms delay time in timer
TMR1L = 0x20;
TMR1H = 0xD1;
count ++;
if (count >= 1000) //measure upto 1000 ms i.e. 1 seconds
{
Buzzer = ~Buzzer;
/* Toggle buzzer pin     */
count = 0;     //reset count
}
TMR1IF = 0; //timer1 overflow flag to 0
}
}
void main()
{
TRISB=0;
/* Set as o
```

```
utput port */
TRISAbits.TRISA5 = 0;            //set buzzer pin RA5 as output
GIE=1;                                /* Enable Global Interrupt */
PEIE=1;                               /* Enable Peripheral Interrupt */
TMR1IE=1;                              /* Enable Timer1 Ov
erflow Interrupt */
TMR1IF=0;
/* Enable 16
-
bit TMR1 register,no pre
-
scale,internal clock, timer OFF */
T1CON=0x80;                 /*1:8 prescale*/
TMR1L = 0x20;
TMR1H = 0xD1;
TMR1ON=1;
/* Turn ON Timer1 */
while(1);
}
```

output:-


File Edit View Tool Design Graph Debug Library Template System Help
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Schematic Capture
Regards : ASIF IQBAL
24v
BUZ1(1)
BUZ1
6v
BUZZER
R1
1k
Q1
BC547
GRAPHICS
COMPONENT
PIN
PORT
MARKER
ACTUATOR
INDICATOR
VPROBE
IPROBE
TAPE
GENERATOR
TERMINAL
SUBCIRCUIT
2D GRAPHIC
WIRE DOT
WIRE
BUS WIRE
BORDER
TEMPLATE
2 Message(s)
ANIMATING: 00:00:01.800000 (CPU load 1%)
+300.0
+500.0